



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - ATSB-style
final report

Drafted 02/10/2026

INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

On 18/06/2026 at 17:30 local time (07:30 UTC), Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off from runway 09 at Tarlton Springs Aerodrome. It was the flight crew's fifth sector of the duty, flown in full darkness after last light. No runway inspection had been carried out since before the first arrival that morning, and none was requested.

At 116 kt, 2 seconds after V1, the first officer saw two kangaroos crossing the runway from the northern side. The pilot in command rejected the take-off at 118 kt, 6 kt above V1, with 695 m of sealed surface remaining. One second later the No. 2 propeller and right main gear struck both animals, and No. 2 reverse thrust became unavailable. The aircraft left the runway end at about 21 kt. It stopped upright on the extended centreline, 62 m into the 90 m runway end safety area (RESA) and 28 m short of ungraded terrain.

None of the 44 passengers and 3 crew was seriously injured. The cabin crew member reported bruising during disembarkation by the forward airstair, and there was no evacuation. The aircraft was substantially damaged:

- three No. 2 propeller blades were damaged beyond limits
- both right main tyres deflated
- the right main gear doors were distorted
- the lower right fuselage was punctured.

Both kangaroos were killed, and the aerodrome was closed for 4 hours 20 minutes.

What the investigation found

The stop began 6 kt late against a scheduled margin of only 67 m. The strike then removed half the reverse thrust, so the achieved deceleration was approximately 15% below the certified value.

The operator required a runway inspection before the first departure after last light only at aerodromes listed in its Appendix 8A. The only listing criterion was a reported company wildlife strike, so the aerodrome's log of 11 kangaroo sightings on the sealed surface could not place Tarlton Springs on the list. This very likely contributed. The aerodrome operator had deferred repair of a failed northern perimeter fence for three budget cycles, leaving an animal pad through the fence on the side of the strike. This also very likely contributed. The operator introduced the evening service as a schedule change without reopening a route risk assessment that restricted the aerodrome to daylight operations, and this likely contributed.

The operator's rejected take-off training covered only aircraft system failures, with no guidance on a runway obstruction at any speed. This likely contributed by leaving the pilot in command without a practised frame for the decision. He also stated that he knew the V1 rule and departed from it deliberately, and on that reading the gap is a finding about the fleet rather than a cause of this reject. The aviation standard required a documented wildlife plan only above a size threshold the aerodrome did not meet. This likely contributed, although on a finer balance than the others.

Partially unserviceable floor proximity emergency lighting did not contribute but increased risk to occupants, because no routine check could detect the fault. The crew's long, late duty also increased risk but was not found to have affected their actions. Whether runway edge lighting intensity affected detection of the animals could not be determined. A tailwind and an engine power loss before the reject were ruled out. The pilot in command and first officer achieved

maximum braking 2 seconds after the decision, faster than the fleet mean of 3.2 seconds, which limited the severity of the overrun.

What has been done as a result

On 25/06/2026 the aerodrome operator installed a temporary electric wire along the northern boundary, funded the fence repair and programmed fence replacement for 09/2026. On 26/06/2026 the operator added Tarlton Springs to Appendix 8A. The investigation recommended that the operator list aerodromes based on wildlife exposure and trigger a route risk assessment for any aerodrome first operated into after last light. A response was due by 30/09/2026, and the recommendation is open. No safety action has been recorded for the size threshold, the change approval, the training gap or the emergency lighting.

Safety message

A defence armed only after a strike cannot prevent the first one. Operators should identify wildlife exposure from sighting information aerodromes already hold, and reassess routes whenever operations move into darkness. At aerodromes with known wildlife activity, crews can request a runway inspection before departing after last light, even where no procedure requires one.

The occurrence

Before departure

On 18/06/2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was scheduled to fly from Tarlton Springs Aerodrome to Sydney. Times are UTC unless stated otherwise. The crew's duty record placed the occurrence at 17:30 local time (07:30 UTC).

The aerodrome operator's daily movement area inspection, carried out at 21:35 on 17/06/2026 before the first scheduled arrival, recorded the movement area as clear. No further runway inspection was carried out or requested that day.

SB512 was the last scheduled sector of the day and the fifth of the flight crew's duty, added that morning when the rostered crew reported unfit. The doors closed at 07:16, 40 minutes behind schedule, with 44 passengers, 2 flight crew and 1 cabin crew on board. The first officer was the pilot flying. The calculated V1 was 112 kt for a weight of 20,800 kg, flap 10, a dry runway 09 and a 5 kt headwind component.

The occurrence notification recorded that "The aerodrome was in darkness" at departure. [NOT ESTABLISHED: the time of last light or end of civil twilight at Tarlton Springs on 18/06/2026 relative to the departure.]

At 07:29:04 the pilot in command reported ready for runway 09, and at 07:29:10 the tower cleared SB512 for take-off with the wind 090 at 6 kt. No runway inspection was offered or requested, and no wildlife transmission was found on the tower channel in the preceding 90 minutes. The pilot in command then remarked "Right, clearance is in, lights are on. Your aeroplane, your take-off."

The take-off and rejected take-off

The first officer began setting power at 07:29:41 and called "Power set" at 07:30:02. The flight data recorder (FDR) showed both power levers advanced together, with torque of 98.4% on the No. 1 engine and 98.5% on the No. 2 engine. The pilot in command responded "Power set, both engines good." Both engines remained at take-off torque, within 0.1% of each other, until the reject, with no exceedance, caution or warning. The headwind component was 4 to 6 kt throughout the roll.

The 80 kt cross-check was made at 07:30:15, and the pilot in command called "V one" at 07:30:29 at 112 kt indicated airspeed.

At 07:30:31, at 116 kt, the first officer called "Roos - roos, roos, on the left, they're on the -". She later described the animals coming out of the dark on the left of the strip and crossing the centreline, in sight for about two seconds or less, and estimated the range at first sight as around 200 m, acknowledging it could have been less. The occurrence notification recorded the two kangaroos moving onto the runway from the northern side, past the 1,100 m point.

The pilot in command's accounts differ. Two minutes after the aircraft stopped, he told the first officer "I never had them. I was heads-in on the engine page after V one." In his air safety report the next day, he wrote that he looked up after the first officer's call and saw two large animals already on the sealed surface, moving left to right across the centreline. The investigation did not reconcile these accounts.

At 07:30:32 the pilot in command called "Stopping. I have control." The first officer responded "You have control." The reject was initiated at 118 kt, 6 kt and 3 seconds after V1, and both power levers reached flight idle within 0.6 seconds. Reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09, with 695 m of sealed surface remaining. In his air safety report

the pilot in command described judging that continuing would have put the aircraft airborne at night with damage that could not be assessed, over country offering no usable option.

At 07:30:33 the aircraft struck both animals. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The cockpit voice recorder (CVR) captured a sound consistent with a heavy impact, lasting about 0.4 seconds, followed by airframe rumble. The right main landing gear and the No. 2 propeller took the impact. At 07:30:34 the first officer reported that they had taken one on the propeller.

Maximum braking was established at 07:30:34, at 109 kt and 2 seconds after the decision, at the fast end of the operator's recorded fleet distribution for rejected take-offs (mean 3.2 seconds). Anti-skid was active on all four wheels and brake pressure was 2,940 psi. At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse was selected on the No. 1 engine only. No. 2 propeller RPM decayed from 07:30:33, consistent with the blade damage later found, and the No. 2 reverse interlock did not make.

Between 07:30:34 and 07:30:44 the mean deceleration was 2.6 m/s², approximately 15% below the certified accelerate-stop value for that weight and runway condition, which assumes symmetric reverse. The first officer called the speed decreasing at 07:30:41. At 07:30:47 the pilot in command said "We're not stopping on it," and at 07:30:49 the first officer called "End's coming up."

At 07:30:51 the aircraft departed the end of runway 09 at about 21 kt indicated airspeed and 20 kt groundspeed. In the graded gravel the longitudinal deceleration fell to 1.9 m/s². The aircraft came to rest at 07:30:57, upright on the extended centreline, 62 m into the 90 m runway end safety area (RESA) and 28 m short of its end, beyond which the terrain is ungraded. The stopping distance from 118 kt was 757 m.

After the aircraft stopped

At 07:30:59 the pilot in command checked that the first officer was uninjured. At 07:31:04 he stated "Parking brake set. Engines - both to cut-off, no fire indications, nothing on the board. I'm not evacuating." The first officer agreed. Both engines were shut down at 07:31:26, and at 07:31:34 the pilot in command told the cabin to remain seated, advising that the aircraft had stopped on the aerodrome and there was no fire.

When the tower called at 07:31:48, the pilot in command reported the reject and runway excursion, 44 passengers and three crew on board, no injuries reported and no fire, and requested activation of the emergency plan and a vehicle. At 07:32:10 the tower activated the aerodrome emergency plan and dispatched a vehicle. Aerodrome staff attended.

At 07:33:02 the pilot in command briefed the cabin that passengers would disembark by the forward door onto the ground, "nice and slow, no rush, no evacuation." No slides were deployed, and the main cabin lights were on throughout. The CVR captured a sound consistent with the forward airstair operating at 07:34:40. The occurrence notification recorded that passengers were disembarked 4 minutes 8 seconds after the aircraft stopped. The timeline instead places the disembarkation from 07:34:40 to 07:38:48, an interval of 4 minutes 8 seconds that began almost 4 minutes after the aircraft stopped. During the disembarkation the first officer noticed unlit sections of the cabin floor proximity strip aft. She raised this with the engineer the next morning, and a functional check was carried out on 24/06/2026.

No serious injuries occurred. The cabin crew member later reported bruising sustained during the disembarkation. Both kangaroos were killed, their carcasses recovered 1,188 m and 1,221 m from the runway 09 threshold. Continuous tyre marks ran from 1,160 m to the runway end. The aircraft was substantially damaged:

- three No. 2 propeller blades were damaged beyond repairable limits, and the engine required a shock-load inspection
- both right main tyres deflated through the fuse plugs, with brake energy absorbed calculated at 84% of the certified maximum
- the right main gear doors were distorted
- the lower right fuselage skin was abraded and punctured in two places.

Both flight crew were stood down and tested for drugs and alcohol, with negative results. The aerodrome was closed for 4 hours 20 minutes, and the FDR and CVR were removed at 12:40.

Context

Flight crew

The pilot in command held 8,410 hours total aeronautical experience, including 2,140 hours on type, and had been in command for 3 years 1 month. His last cyclic simulator session (11/03/2026) and last line check (02/12/2025) were satisfactory. He had no prior occurrence history. His duty and rest in the preceding 7 days were within the operator's limits, with no variations applied.

The first officer held 1,655 hours total, including 405 hours on type, and had been with the operator for 14 months. Her last cyclic simulator session (28/04/2026) was satisfactory. She had flown two previous daylight sectors into Tarlton Springs, on 09/01/2026 and 17/02/2026. Before the flight she reviewed the flight plan, charts, ERSA entry and the operator's aerodrome brief page. None referred to wildlife at Tarlton Springs.

Duty

The flight crew signed on at 06:30 local time. SB512 was the fifth sector of a five-sector duty, added that morning when the rostered crew reported unfit. The occurrence happened 11 hours into a duty planned to finish at 18:35 local, a 12 hour 05 minute duty against a maximum of 12 hours 30 minutes. The departure was 40 minutes late. Both flight crew described themselves as tired but not struggling. The pilot in command stated that he would have declined a further sector.

Aircraft

VH-SBQ was a 50-seat twin-turboprop operated by Saltbush Air. Aerodrome Standard information gave the type's weight as 22,900 kg. The take-off weight for SB512 was 20,800 kg.

Post-occurrence borescope inspection of both engines found no pre-existing damage and no evidence of a power anomaly before the impact. The anti-skid system tested serviceable. The right main landing gear structure was within limits.

Cabin emergency lighting

After the first officer reported unlit floor strip sections during the disembarkation, a functional check of the cabin emergency lighting was carried out on 24/06/2026. With the system armed on battery power, two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate, owing to a failed ballast common to both.

The type's pre-flight emergency lighting test neither illuminates nor monitors the floor proximity segments, so the defect could not be detected by the flight crew or a line engineer without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. The disembarkation was conducted by the forward airstair, with the main cabin lights on and without urgency.

Meteorology and light conditions

The routine observations for Tarlton Springs (YTLS) around the departure were:

- "METAR YTLS 180630Z 09007KT 9999 FEW040 09/M01 Q1024" [weather observations]
- "METAR YTLS 180700Z 09006KT 9999 FEW040 08/M01 Q1024" [weather observations]
- "METAR YTLS 180800Z 09005KT 9999 FEW040 07/M02 Q1024" [weather observations]

These gave a headwind component of 5 to 7 kt on runway 09. The crew's performance calculation used 5 kt. The crew's forecast, TAF YTLS 180458Z, gave 09008KT 9999 FEW040, becoming 08006KT CAVOK from 10:00. Runway 09 was dry.

On 18/06/2026 (AEST, UTC+10), sunset at Tarlton Springs was at 16:58 and civil twilight ended at 17:25. The take-off roll began at about 17:30 local time. Moonrise was at 22:41, with a waning crescent moon 14% illuminated. There was no natural ambient light on the runway strip and no town lighting within sight. A site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip either side of the sealed surface could not be seen from the runway centreline.

Aerodrome

Runway and runway end safety area

Tarlton Springs is a certified aerodrome with a single runway, 09/27. The accelerate-stop distance available (ASDA) on runway 09 was 1,847 m. Beyond the runway 09 end is a 90 m graded gravel RESA, then ungraded terrain. The tower was open, rostered to close at 18:00 local.

Runway edge lighting

A photometric check of the runway 09/27 edge lighting on 25/06/2026, seven days after the occurrence, found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal stage 3 intensity. The remainder were within tolerance. The aerodrome's maintenance programme checked lamp serviceability but not intensity. No measurement exists for the night of the occurrence.

Perimeter fence

Works register item WR-221, raised in 04/2023, described the northern boundary fence between the 700 m and 1,300 m markers as having four displaced panels and strained wire, with animals observed passing through. Replacement was recommended at an estimated \$41,000 and deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, recorded further deterioration.

On 23/06/2026 the investigation found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts, and a clear animal pad through it at the 1,050 m mark. This was on the same side of the runway as, and 138 m before, the first carcass recovered at 1,188 m (photograph: attachments/Perimeter_fence_repair_deferred_for_three_budget_cycles_1.jpg).

Wildlife sighting log

The aerodrome kept a paper wildlife sighting log. Between 01/2025 and 06/2026 it held 46 entries, 41 recording kangaroos on or adjacent to the movement area. Of those, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was not reported to any operator, and the aerodrome reporting officer confirmed on 23/06/2026 that no airline had ever asked for it.

The pilot in command had seen kangaroos on the strip at Tarlton Springs in the evening on several previous occasions. He had mentioned this to ground staff but had not reported it to the operator.

Runway inspections

Aerodrome Manual 6.3 provides for a scheduled inspection before the first arrival and further inspections at the request of the reporting officer or a pilot in command. The reporting officer confirmed that an inspection would have been carried out if requested.

Wildlife hazard management requirements

Aerodrome Standard AS-139 section 10.4 states:

- "10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." [AS-139]
- "10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year." [AS-139]
- "10.4.3 Where subsection 10.4.2 does not apply, the measures taken under 10.4.1 need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place." [AS-139]

The heaviest scheduled aeroplane was the 22,900 kg twin-turboprop, and the aerodrome handled 61,400 passenger movements in 2025, so section 10.4.2 did not apply. The aerodrome held no wildlife hazard management plan, hazard assessment or dispersal programme, and no documented measures beyond the pre-first-arrival inspection. It was not in breach of the standard.

Operator procedures

Wildlife procedures

Operations Manual Part A section 8.14 (revision 9, 04/2024) states that wildlife "presents a hazard to aircraft operations, particularly at regional aerodromes and particularly around dawn and dusk." Section 8.14.2 requires that, at an aerodrome listed in Appendix 8A, "the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." Section 8.14.3 states that elsewhere "no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area."

Appendix 8A states that "An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Section 8.14.4 requires a strike to be reported on an air safety report and the aerodrome's status reviewed. Part A had no provision for reporting sightings and no other route to listing. On 18/06/2026 Appendix 8A listed nine aerodromes, last reviewed on 03/03/2025. Tarlton Springs had no reported company strike and had never been listed.

Rejected take-off procedures

Operations Manual Part B section 4.6 (revision 14, 09/2025) sets out the rejected take-off criteria. Between 80 kt and V1, section 4.6.3 provides that the take-off "shall be rejected only for: fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly." Section 4.6.4 states "At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached." Section 4.6 contained no guidance on a runway obstruction at any speed, and no amplification of 4.6.4 for a case in which continuing is itself hazardous.

Rejected take-off training

Module 4 of recurrent training syllabus TRG-SYL-07 (cycle 2025 to 2026) contained four scripted simulator scenarios: an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt, continued. All were aircraft system events, three below V1, and none presented an external hazard on the runway. In that cycle 214 crew completed module 4. No crew had been given an external obstruction scenario in the five training cycles on file. Both flight crew stated independently that every stop or go scenario they had been given was an engine event below V1.

Route risk assessment and the evening service

The operator's route risk assessment for Tarlton Springs, RRA-041 issue 3 (11/2022), assessed runway length and accelerate-stop margin, a single runway with no lighting redundancy, no alternate within 200 nm at night, limited engineering support, and summer heat and density altitude. It carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy." Issue 3 was current and no issue 4 existed.

Change Management Record CMR-2025-118, approved by the Head of Flying Operations on 22/05/2025, introduced an evening Sydney to Tarlton Springs rotation, six days per week, from 14/07/2025. This was the operator's first scheduled operation there after last light. The change was assessed as "schedule change only" with "No new hazard identified", and "No" was answered to "Route risk assessment review required". It did not refer to the daylight-only restriction, and RRA-041 was not reopened. The Safety Department only received the approved change. The approver had left the company and could not be interviewed. The pilot in command stated that no briefing or other change accompanying the evening service had reached him.

Performance data

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. It assumed a reject begun at or before V1 with reverse thrust available on both engines.

Safety analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4 sets out two levels of obligation for wildlife hazard management at a certified aerodrome. Every operator of a certified aerodrome "must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome" (section 10.4.1). A documented wildlife hazard management plan is required only "where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year" (section 10.4.2). Where neither limb applies, section 10.4.3 provides that the measures "need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place."

Tarlton Springs fell below both limbs. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. The aerodrome therefore held no wildlife hazard management plan. It also held no hazard assessment and no dispersal programme. Apart from the single pre-first-arrival inspection under Aerodrome Manual 6.3, it had no documented measures. This position was confirmed against the certified aerodrome register and the aerodrome operator's own compliance file, and the aerodrome was not in breach of the standard. Because the standard's text and the aerodrome's traffic figures are documentary and not in dispute, it is virtually certain that the threshold in section 10.4.2 left Tarlton Springs without any requirement for a documented plan.

The aerodrome was not short of knowledge about the hazard. The wildlife sighting log for January 2025 to June 2026 held 46 entries:

- 41 of them recorded kangaroos on or adjacent to the movement area.
- 38 of those 41 were timed between one hour before and two hours after last light.
- Eleven entries recorded animals on the sealed surface of runway 09/27.

The log was a paper register held in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked to see it. The knowledge that kangaroo activity peaked around last light, and that animals reached the runway seal, therefore existed at the aerodrome but did not reach the operator that scheduled a departure into that period.

The aerodrome works register showed the same pattern for the perimeter fence. Item WR-221 was raised in 04/2023. It recorded displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, noted that animals had been seen passing through, and recommended replacement at an estimated \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, recorded further deterioration of the same section. The site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

The argument that the threshold influenced the occurrence rests on what a documented plan provides. A documented plan has an owner and an audit trail. It is the mechanism that would have connected the aerodrome's sighting log to the operators using the aerodrome. It would also have placed the failed fence on a funded programme rather than leaving it as an unfunded entry on a

works register. Without a plan, the sighting log and the fence defect each sat with nobody required to act on them.

There was also evidence against this conclusion. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures whether or not a plan was documented. A documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The council's three deferrals were budget decisions, and there was no evidence that a documented plan would have attracted funding the works register did not. A plan sets out what should be done, but the fence repair depended on money being allocated.

The investigation weighed these points and concluded that the threshold likely contributed to the occurrence. The deciding consideration was that the controls that did not work in this occurrence did not work for want of an owner and an audit trail, which is what a documented plan supplies. This conclusion is less firm than the investigation's conclusions on the other contributing factors. It is reached on the balance of the evidence rather than settled, and the evidence that a plan would not have changed the funding outcome for the fence was given weight in reaching it.

Two contributing factors followed from this condition, and each is argued in its own subsection. The deferral of the perimeter fence repair over three budget cycles very likely contributed to the animals' entry onto the runway strip. The operator's arrangement requiring a runway inspection only at listed aerodromes, with a strike as the listing criterion, very likely contributed to no inspection being requested before the departure. This subsection is concerned only with the external requirement that left an aerodrome of this size with an undocumented and unaudited wildlife programme. In practice that programme was also unfunded, while the aerodrome remained fully compliant with the standard.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The aerodrome operator's works register showed that the condition of the northern perimeter fence was known well before the occurrence. Item WR-221 was raised in 04/2023. It recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, noted that animals had been observed passing through, and recommended replacement at an estimated \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time as unfunded and for review at the next cycle. Item WR-263, raised in 02/2026, recorded the same section as deteriorating further.

It was virtually certain that the repair had been deferred in this way. The deferrals were set out in the aerodrome operator's own register in its own words, and nothing had to be reconstructed. The site inspection on 23/06/2026 confirmed the condition the register described. It found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts, and a clear animal pad through it at the 1,050 m mark (Figure 1).



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The aerodrome's wildlife sighting log for 01/2025 to 06/2026 showed what the fence was failing to exclude. Of 46 entries, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and 11 recorded animals on the sealed surface of runway 09/27. The log was a paper register held in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it. The aerodrome had therefore documented the exposure, but that knowledge did not reach the operators flying there.

The physical evidence connected the breach to the strike. The animal pad crossed the fence at the 1,050 m mark on the northern side. The first officer saw the two kangaroos moving onto the runway from the northern side, past the 1,100 m point. The carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold. A fence in the condition found does not exclude kangaroos, and the pad through the breach was directly upstream of the strike position on the same side of the runway. It was very likely that the deferred repair contributed to the two kangaroos entering the runway strip and moving onto the sealed surface.

The deferral occurred within the regulatory arrangement discussed under External Influences, under which the aerodrome was not required to hold a documented wildlife hazard management plan. The timing of the departure, in the hour after last light, was a separate contributing condition and is discussed with the local conditions.

Evening service approved as a schedule change, route risk assessment not reopened

At the time of the occurrence, the operator's current route risk assessment for Tarlton Springs was RRA-041 issue 3, dated 11/2022. It assessed hazards including runway length and accelerate-stop margin, a "single runway with no lighting redundancy" and "no alternate within 200 nm at night". It carried the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy." Issue 3 was the current issue and no issue 4 existed.

Change Management Record CMR-2025-118 introduced "one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025." This was the first scheduled company operation into Tarlton Springs after last light. The record assessed the change as "schedule change only. Existing aerodrome, existing aircraft type, existing crew qualifications, existing engineering support arrangements. No new hazard identified." It answered "No" to whether a route risk assessment review was required. The Head of Flying Operations approved it on 22/05/2025. The daylight-only restriction was not referenced anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records showed no involvement in the change beyond receipt of the approved record.

It was virtually certain that the evening service was approved in this way and that the route risk assessment was not reopened. Each document was a current controlled copy, the document register held no later issue of RRA-041, and nothing contradicted either document. The assessment asked whether the elements of the change were new, and each element it listed already existed. It did not ask whether the existing assessment still held for an operation into darkness. The current restriction addressed exactly that question.

The approver had left the company and could not be interviewed. The reasons the change was assessed as a schedule change only could not be established, and the investigation drew no conclusion about intent.

The operator's Operations Manual Part A identified wildlife as a hazard "particularly at regional aerodromes and particularly around dawn and dusk" (section 8.14.1). A review of RRA-041 for a night operation would very plausibly have reached the aerodrome's wildlife exposure and its absence from Appendix 8A in the same pass. Two considerations weighed against a stronger conclusion. First, the restriction named only night lighting and diversion fuel policy, so a review might have addressed those matters alone. Second, the approver could not be asked what information was in front of him at the time. On that basis, it was likely that approving the evening service without reopening the route risk assessment contributed to the absence of any requirement for a runway inspection before the first departure after last light.

The listing criterion for Appendix 8A, and the absence of a runway inspection request on the night, are discussed with the risk controls and individual actions respectively.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator had a defence against wildlife on the runway at night. Operations Manual Part A 8.14.2 states that at an aerodrome listed in Appendix 8A "the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." For every other aerodrome, 8.14.3 states that "no specific action is required."

Appendix 8A lists an aerodrome only "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nine aerodromes were listed on the day of the occurrence. Tarlton Springs was not one of them and had never been listed. No provision

in Part A asked a crew to report a sighting, and none allowed an aerodrome onto the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.

It was virtually certain that this condition existed, because both the manual text and the appendix listing are documents. The weakness lay in what armed the control. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 held 41 entries recording kangaroos on or adjacent to the movement area. Of these, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it. None of those entries could put Tarlton Springs on the list, because none was a strike involving a company aircraft. The only way onto the list was the kind of event the inspection existed to prevent.

Had the requirement applied, an inspection would have been requested and a result passed before the roll, and aerodrome staff disperse animals when they find them. It was very likely that the absence of this requirement at Tarlton Springs contributed to the absence of an inspection on the night. This is not rated higher because an inspection completed some minutes before departure does not guarantee a clear runway when the roll begins. The crew's part, the absence of a request before the first departure after last light, is discussed under Individual Actions. How the aerodrome came to be outside any documented wildlife arrangement is discussed under External Influences.

Rejected take-off training covers aircraft system failures below V1 only

The objective of recurrent training syllabus TRG-SYL-07 module 4 is a rejected take-off "for an engine failure, an engine fire and a configuration warning, each initiated below V1," and "the decision to continue for a failure recognised at or after V1." Its four scripted scenarios are engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt and engine failure at V1 plus 5 kt, continued. All four are aircraft system events. In the five training cycles on file, no crew was presented with an external obstruction, and 214 crew completed the module in the 2025 to 2026 cycle alone. Operations Manual Part B 4.6 is consistent with this. Paragraph 4.6.4 states that "at or above V1 the take-off shall be continued," and the section gives no guidance on an obstruction on the runway at any speed. It was extremely likely that this condition existed. It is not rated virtually certain only because scenario records from before the five cycles on file were not available. Both flight crew stated independently that every stop or go scenario they had been given was an engine event below V1.

The evidence supported two readings of how this gap bore on the decision. On the first, the captain had about two seconds and no prepared frame for the situation. The decision to reject was the first time he had considered the question, which is what training is meant to prevent. On the second, the captain stated plainly that he knew the rule and departed from it deliberately because he judged continuing to be worse. On that reading, training on obstructions would not have changed what he knew, and the gap is a matter for the fleet rather than a cause of this rejected take-off.

The investigation found it likely that the training gap contributed to the take-off being rejected 6 kt above V1. An unrehearsed decision made in two seconds is not the same as one made against a briefed and practised frame. However, the second reading was not ruled out, and the finding is limited to likely for that reason. The gap explains the conditions under which the decision was made. It does not imply that the decision would have been different, and the decision itself is discussed under Individual Actions.

Cabin floor proximity emergency lighting partially unserviceable

During the disembarkation, the first officer noticed unlit sections of the aisle floor strip. A functional check on 24/06/2026 found that two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate on battery power with the system armed. The fault was traced to a failed ballast common to both segments. It was virtually certain that this condition existed.

It was unlikely that the defect contributed to the occurrence or its outcome. No evacuation was commanded. The main cabin lights were on throughout, and passengers left by the forward airstair without urgency, 4 minutes 8 seconds after the aircraft stopped.

It was very likely that the defect increased risk nevertheless. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but neither illuminates nor monitors the floor segments. The defect therefore could not be detected by the crew or by a line engineer without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. The same ballast is fitted across the type, so the condition of the escape path lighting in the rest of the fleet was not known. In an evacuation at night with smoke, or over the wing, two dark segments in the aft cabin could make the difference between an orderly exit and an obstructed one.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local time. That was 32 minutes after sunset at 16:58 and 5 minutes after the end of civil twilight at 17:25. Moonrise was not until 22:41, and the moon was a waning crescent, 14% illuminated. The aerodrome was in full darkness, with no natural ambient light on the strip and no town lighting within sight. A site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip on either side of the sealed surface could not be seen from the runway centreline.

The aerodrome's own wildlife sighting log covered January 2025 to June 2026 and held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The astronomical data and the log are documentary. However, the log is a paper register and its completeness could not be verified. For that reason, the departure in this period of darkness and peak wildlife movement was assessed as extremely likely rather than certain.

The first officer described the animals in a way that matched this light condition. On the cockpit voice recorder she said, "They came out of the dark on the left of the strip, crossed the centreline." She added, "There was nothing to see and then there were two of them." The animals were not seen until they were on the lit surface, and the departure coincided with the recorded movement peak. It was therefore likely that this condition contributed to the two kangaroos entering the runway strip and moving onto the sealed surface undetected. On its own, a departure after last light was unremarkable. It mattered because no runway inspection was requested before the first departure after last light. That absence was found to be a contributing factor, and the inspection requirement is discussed under Risk Controls.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the intensity of the runway edge lighting reduced the crew's ability to see the animals. This matter could not be resolved.

Two pieces of evidence supported the proposition. First, the pilot in command recalled the lighting as brighter than he would have wanted. He described it as a corridor of light with everything outside it black, and said that nothing off the seal could be seen coming. Second, a photometric check on 25/06/2026 found four edge lights on the northern side, between the 900 m and 1,200

m markers, at 118% to 131% of nominal intensity for stage 3. This was the section the animals crossed. The aerodrome operator's maintenance programme checked lamp serviceability but did not measure intensity. The deviation in intensity is established. Whether that deviation reduced the crew's ability to see an animal on the runway 09 surface is an inference, and the inference is what was examined.

Other evidence weighed against it. The first officer was the crew member who actually saw the animals. She described the lighting as normal and said she saw them completely clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge whether the lighting was unusual. The captain's not seeing the animals has an explanation that does not depend on the lighting. He said, "I never had them. I was heads-in on the engine page after V one." That is where the procedure placed his attention after V1.

The matter could not be settled for four reasons:

- The photometry was taken seven days after the occurrence.
- Nothing records what the lights were doing on the night.
- The two crew accounts disagree.
- No study links an edge light at 131% of nominal to the detection of an object on the seal.

Resolving the matter would require a measurement of the lights as they were on the night and evidence connecting that intensity to the detection of an object on the runway. Neither exists. The proposition was assessed as about as likely as not. It remains unresolved and is not carried into the findings.

Take-off commenced with a tailwind above limits

A tailwind was ruled out as very unlikely. The observations at 06:30, 07:00 and 08:00 UTC ("09007KT", "09006KT", "09005KT"), the recorded aircraft data showing a headwind component of 4 to 6 kt throughout the roll, and the crew's 5 kt headwind performance entry all agreed that there was no tailwind at any point.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

The crew signed on at 06:30 local time for five sectors. The fifth sector was added that morning when the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into a duty planned to finish at 18:35. That gave a 12 hour 05 minute duty against a limit of 12 hours 30 minutes for five sectors from that sign-on. The aircraft departed 40 minutes behind schedule. These circumstances were virtually certain to have existed.

The argument for an effect on crew performance was that 11 hours into a five-sector day is within the range where degraded decision making is expected. The schedule was also late, and the crew were working to finish. The recorded evidence did not support any mechanism through which such an effect could have operated:

- The power levers reached flight idle 0.6 seconds after the decision.
- Maximum braking was established 2 seconds after the decision, against a fleet mean of 3.2 seconds.
- There was no error of omission during the roll.
- The one decision the schedule might plausibly have affected was the request for a runway inspection. That decision was never available to be affected, because no document prompted either crew member to consider it.

Both crew described themselves in interview as tired but not struggling. The captain said that he would have declined a further sector and that he was fully attentive at the critical moment. It was therefore unlikely that the duty contributed to the occurrence.

The duty pattern was nevertheless likely to have increased risk. The evening Tarlton Springs rotation routinely placed the last sector within the last hour of a 12-hour duty. That sector operated to an aerodrome with no company engineering, no alternate within 200 nm at night, and no crewing resilience. As a result, an unfit crew member could turn a four-sector day into a five-sector day on the morning of operation. This is a standing exposure built into the schedule and not a one-day event.

Individual Actions

No runway inspection requested before the first departure after last light

No runway inspection was requested or offered before SB512 departed. Neither the CVR nor the tower's recorded audio contains a request for an inspection, an offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 shows a single inspection, at 07:35 local before the first arrival, with the result 'movement area clear'. No later inspection was carried out or requested. Taken together, these two independent recordings and the aerodrome's own log mean it is virtually certain that no inspection took place.

The inspection was available. Aerodrome Manual 6.3 provides for inspections outside the scheduled time when the reporting officer or a pilot in command asks for one. The reporting officer confirmed that an inspection would have been carried out if it had been requested.

Neither crew member considered asking. The captain had expected Tarlton Springs to be on the operator's Appendix 8A list and only found out afterwards that it was not. Nothing in the operator's documents prompted either crew member to request an inspection at this aerodrome. The way the listing criterion kept Tarlton Springs off that list is discussed under Risk Controls. Seen this way, the omission had no owner. No procedure required the crew to consider an inspection, and the aerodrome would have done one if asked.

An inspection would have covered the section from 1,050 m to 1,250 m, where the fence pad and both carcasses were found. Aerodrome staff disperse animals when they find them. It was therefore likely that the absence of an inspection contributed to the animals being on the sealed surface during the roll. This is not put higher because an inspection made some minutes before the take-off roll cannot guarantee that the runway stays clear.

Take-off rejected 6 kt above V1

V1 was called at 112 kt at 07:30:29. The first officer's sighting call came at 07:30:31, when the aircraft was at 116 kt and no control input had yet been made. The CVR records the decision and the handover of control in a single exchange:

- "Roos - roos, roos, on the left, they're on the -" (first officer, 07:30:31)
- "Stopping. I have control." (captain, 07:30:32)

The recorded data shows both power levers moving to flight idle from 07:30:32 at 118 kt, which was 3 seconds and 6 kt after V1. The captain later stated on the CVR, "I never had them. I was heads-in on the engine page after V one." He therefore made the decision on the first officer's call alone. In the first officer's words, the animals had been in view for "two seconds. Maybe less."

Part B 4.6.4 requires the take-off to be continued at or above V1. Section 4.6 contains no guidance on an obstruction seen on the runway at any speed. It also says nothing about a case in which continuing is itself hazardous. In interview and in his air safety report, the captain explained his

reasoning. He judged that rotating into two large animals would leave the aircraft airborne at night, with damage he could not assess, over country that offered no usable option. He chose the risk whose end he could see. He stated plainly that the decision was not in accordance with the procedure.

It is virtually certain that the reject occurred as described. The FDR and CVR establish it, and the captain's own account corroborates it. The aircraft was 1,152 m along runway 09 when the reject began, which left 695 m of sealed surface. The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1. The margin between the accelerate-stop distance required and the distance available was 67 m. Beginning the stop 6 kt and 3 seconds late used up that margin and more, even before the loss of No. 2 reverse is taken into account. It was extremely likely that the late reject contributed to the reduced stopping performance and the overrun. The aircraft's achieved stopping distance from 118 kt was 757 m.

The decision did not arise in isolation. It followed from the animals being on the sealed surface at all. It was also shaped by rejected take-off training that covers only aircraft system failures below V1, which is discussed under Risk Controls. Whether the runway edge lighting intensity affected how early the animals could be detected was discussed under Local Conditions and could not be resolved. Nothing from that matter is carried into this analysis.

Maximum braking and No. 1 reverse applied without delay

The recorded data shows the stop was carried out quickly once the decision was made:

- The power levers reached flight idle 0.6 seconds after the decision.
- Maximum braking was established at 07:30:34, 2 seconds after the decision. The operator's recorded fleet mean from decision to maximum braking is 3.2 seconds, so this response was at the fast end of the fleet distribution.
- Anti-skid was active on all four wheels throughout the stop.
- Brake energy reached 84% of the certified maximum, without a brake fire.
- The captain selected No. 1 reverse at 07:30:36, 4 seconds after the decision. No. 2 reverse was unavailable because its propeller had been destroyed in the strike.

The first officer made the speed callouts and the runway-end call. The aircraft left the seal at 21 kt and stopped 62 m into the 90 m RESA. It came to rest upright, on the extended centreline, 28 m short of the end of the prepared area. It is virtually certain that this response occurred as recorded.

The investigation estimated that a stop begun at the fleet mean response time would have carried the aircraft a further 30 to 40 m. The prompt application of maximum braking and No. 1 reverse very likely reduced the severity of the overrun. Because of it, the aircraft entered a prepared surface at low speed instead of travelling further toward the end of the RESA. The investigation found that the reject was outside procedure. It also found that the actions that followed the reject were executed faster than the operator's fleet average.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

It is virtually certain that two kangaroos entered the runway strip and moved onto the sealed surface of runway 09. Three separate lines of evidence each establish this on their own:

- Two adult kangaroo carcasses were recovered from the northern half of the runway, at 1,188 m and 1,221 m from the runway 09 threshold.
- The CVR recorded the first officer's sighting call at 07:30:31: "Roos - roos, roos, on the left, they're on the -".

- The FDR recorded an impact at 07:30:33.

After the aircraft stopped, the first officer described the animals as having come "out of the dark on the left of the strip" and crossed the centreline. The first officer said they were in view for about "two seconds. Maybe less." That account matches the northern-side position of the carcasses and the direction of travel from north to south. The captain stated "I never had them," and explained that he was looking at the engine page after V1.

The position of the carcasses is consistent with the animals' approach. The site inspection found an animal pad through the failed section of the northern perimeter fence at the 1,050 m mark. The first carcass lay 138 m beyond it, on the same side of the runway. The question of how the animals reached the strip belongs to the analysis of the deferred fence repair under Organisational Influences, and it is not argued here. This finding states only that the animals reached the sealed surface, which is settled.

The presence of animals on the runway at that time of day was not unusual. Over the 18 months before the occurrence, the aerodrome's wildlife sighting log held 41 entries for kangaroos on or near the movement area. Most of these were at or around last light. Eleven entries placed animals on the sealed surface of runway 09/27. The only runway inspection on the day of the occurrence was at 07:35 local, about 10 hours before the departure. The absence of a later inspection is discussed under Individual Actions.

It is extremely likely that the presence of the animals on the runway contributed to the occurrence. The decision to reject the take-off and the strike both followed directly from it. The decision to reject is analysed under Individual Actions. The strike is analysed below.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the aircraft struck both animals, and that the strike contributed to the overrun. The recorded data, the physical damage and the recovered tissue all agree:

- **Recorded data.** The FDR recorded the impact at 07:30:33 at 114 kt, one second after the reject began. It showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The CVR recorded a sound consistent with a heavy impact at the same time.
- **Propeller damage.** Three No. 2 propeller blades were damaged beyond repairable limits, and one blade tip had separated.
- **Animal tissue.** Tissue was recovered from two of the No. 2 propeller blade roots.
- **Gear and fuselage damage.** The right main gear doors were distorted, and the lower right fuselage skin had been punctured by propeller debris.

The strike mattered beyond the damage it caused. No. 2 propeller RPM began to decay from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it was available on No. 1 only. The captain's call at that moment recorded the loss directly: "Reverse - number two's not coming."

The strike therefore removed half of the aircraft's reverse thrust at the start of a stop that was already beyond V1. The scheduled accelerate-stop figures assume symmetric reverse. The effect on the deceleration actually achieved is set out below.

Deceleration achieved below the scheduled accelerate-stop performance

It is virtually certain that the aircraft decelerated at less than the rate assumed in the scheduled accelerate-stop performance, and that this shortfall contributed to the overrun. The recorded data, the distance reconstruction and the physical marks on the runway agree with each other.

The scheduled performance left a margin of 67 m:

- The accelerate-stop distance required was 1,780 m.
- The ASDA was 1,847 m.
- This calculation assumed that the stop began at or before V1, with reverse available on both engines.

Neither assumption held on the night:

- **Late start.** The reject began at 118 kt, 6 kt and 3 seconds after V1. The aircraft was then 1,152 m along runway 09, with 695 m of sealed surface remaining.
- **Reduced deceleration.** With reverse on No. 1 only, the mean deceleration once maximum braking was established was 2.6 m/s². This was approximately 15% below the value used in the certified accelerate-stop data for that weight and runway condition.

The stopping distance actually achieved from 118 kt was 757 m. That is 62 m more than the sealed surface that remained, and it matches the nose wheel rut ending 62 m into the RESA on the extended centreline.

The evidence does not point to any shortfall in the braking system itself:

- Anti-skid was active on all four wheels.
- The system tested serviceable after the occurrence.
- The brakes absorbed 84% of their certified maximum energy, with no fire or hydraulic leak.

The aircraft left the sealed surface at about 21 kt. Its deceleration then fell to 1.9 m/s² in the graded gravel, and it stopped 28 m short of the end of the prepared safety area.

The two departures from the scheduled assumptions acted together. Both the late start and the reduced deceleration were needed to carry the aircraft past the runway end, and neither alone did so. For that reason this shortfall is treated as a single mechanism. The two causal sequences, one through the decision to reject and one through the strike, meet at this point.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss was tested as the trigger for the reject and was ruled out as very unlikely. Both engines remained at take-off torque, within 0.1% of each other, with no caution, warning or exceedance until the power levers were retarded at 07:30:32. The No. 2 RPM decay began only at the impact at 07:30:33.

Findings

From the evidence available, the following findings are made with respect to the rejected take-off and runway overrun involving Saltbush Air SB512, VH-SBQ, at Tarlton Springs Aerodrome on 18/06/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- Aerodrome Standard AS-139 required a documented wildlife hazard management plan only above a size threshold the aerodrome did not meet, and that requirement likely contributed by leaving the aerodrome's wildlife measures without the audit trail and owner a documented plan would have provided. This was reached on a finer balance than the other contributing factors, because section 10.4.1 still required reasonable measures and there was no evidence that a documented plan would have secured funding for the fence. (Safety Issue)
- The aerodrome operator deferred repair of the northern perimeter fence for three budget cycles, and it is very likely that this contributed by leaving a breach through which the kangaroos reached the runway strip on the same side of the runway as the strike. (Safety Issue)
- It is very likely that the departure took place in the hour after last light, the peak of kangaroo movement recorded in the aerodrome's own sighting log, and likely that this contributed, as the animals could not be seen until they were on the lit sealed surface.
- The operator approved the evening service as a schedule change and did not reopen the route risk assessment, despite its daylight-only restriction, and this likely contributed by leaving the aerodrome's wildlife exposure and its absence from Appendix 8A unexamined before night operations began. A review might have addressed only the night lighting and diversion fuel named in the restriction, and the approver could not be interviewed. (Safety Issue)
- With virtual certainty, the operator required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A, and the only criterion for listing was a reported company wildlife strike, so the control could be armed only by the event it was intended to prevent. It is very likely that this contributed, because 11 recorded sightings of kangaroos on the sealed surface at Tarlton Springs could not place it on the list. (Safety Issue)
- It is virtually certain that no runway inspection was requested before the first departure after last light, and likely that this contributed. No procedure prompted either flight crew member to request one, and the aerodrome would have carried out an inspection if asked.
- Two kangaroos entered the runway strip and moved onto the sealed surface of runway 09 in the path of the aircraft, which is virtually certain, and very likely contributed to both the rejected take-off and the strike.
- It is very likely that the operator's rejected take-off training covered aircraft system failures below V1 only, with no external obstruction scenario presented to any crew in the five training cycles on file, and likely that this contributed by leaving the pilot in command to make the reject decision without any briefed or practised frame. The pilot in command also stated that he knew the V1 rule and departed from it deliberately, and on

that reading the training gap is a finding about the fleet rather than a cause of this reject. (Safety Issue)

- The take-off was rejected at 118 kt, 6 kt and 3 seconds above V1, and it is very likely that this contributed by beginning the stop beyond the point assumed in the scheduled accelerate-stop distance, which had a margin of only 67 m.
- With virtual certainty, the No. 2 propeller and right main gear struck two kangaroos 1 second after the reject began, and the strike, again with virtual certainty, contributed by making No. 2 reverse thrust unavailable during the stop.
- The deceleration achieved was below the scheduled accelerate-stop performance, at a mean of 2.6 m/s² (approximately 15% below the certified value), with reverse available on No. 1 only, which is virtually certain, and it virtually certainly contributed, together with the late reject, to the aircraft leaving the runway end at about 21 kt and stopping 62 m into the RESA.

Other factors that increased risk

- It is virtually certain that the cabin floor proximity emergency lighting was partially unserviceable, with two of six escape path segments aft of row 16 on the left side dark because a common ballast had failed. The pre-flight test could not detect the defect, and the 24-month functional test interval could not reliably detect it either, which very likely increased risk to occupants in any evacuation on the type. (Safety Issue)
- The flight crew were operating the fifth sector, 11 hours into duty and 40 minutes behind schedule, after the sector was added that morning when the rostered crew reported unfit, which is virtually certain. This routine exposure on the evening Tarlton Springs rotation likely increased risk.

Other findings

- The take-off was not commenced with a tailwind above limits. The aerodrome observations and the recorded flight data both showed a headwind component throughout the roll.
- There was no No. 2 engine power loss during the take-off roll before the reject. Both engines remained within 0.1% torque of each other until the power levers were retarded, and the No. 2 propeller RPM decay began at the moment of impact.
- Maximum braking and No. 1 reverse were applied without delay, with maximum braking established 2 seconds after the decision against a fleet mean of 3.2 seconds. This reduced risk by limiting the overrun to 62 m into the 90 m RESA at low speed.

Safety issues and actions

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4.2 required a documented wildlife hazard management plan only where an aerodrome was used by aeroplanes above 27,000 kg MTOW or served more than 350,000 passenger movements a year. Tarlton Springs, with a heaviest scheduled aeroplane of 22,900 kg and 61,400 passenger movements in 2025, met neither limb. Under section 10.4.3, its wildlife measures did not need to be documented at all, and the aerodrome was not in breach of the standard.

The issue matters because of what the threshold leaves out. An aerodrome of this size can run a wildlife programme that is undocumented, unaudited and, in practice, unfunded. At Tarlton Springs, the aerodrome's sighting log held 41 kangaroo entries, 11 of them on the sealed surface. It was a paper register that no operator had ever asked to see. The failed northern fence stayed on a works register without funding.

The investigation found that the threshold likely contributed to the occurrence through the deferred fence repair and the operator's runway inspection control. That finding was reached on a finer balance than the other contributing factors. Section 10.4.1 still required reasonable measures, and a documented plan was not the only way to repair a fence or share a sighting log. There was also no evidence that a documented plan would have secured funding for the fence.

No safety action has been recorded in response to this safety issue.

Perimeter fence repair deferred for three budget cycles

The aerodrome operator deferred repair of the northern perimeter fence at Tarlton Springs for three budget cycles. This left a known breach between the 700 m and 1,300 m runway markers open to wildlife.

The defect was first entered in the aerodrome works register as item WR-221 in 04/2023. The entry recorded four displaced panels, strained wire and animals observed passing through, and it estimated replacement at \$41,000. The repair was deferred at the 2023, 2024 and 2025 budget rounds, each entry reading "defer, unfunded, review next cycle". Item WR-263 (02/2026) recorded the same section deteriorating further.

On the site inspection of 23/06/2026, the investigation found a continuous 140 m section of fence with the bottom two wires separated from the posts. There was also an animal pad through the fence at the 1,050 m mark (Figure 1). The carcasses were recovered at 1,188 m and 1,221 m, on the same side of the runway.

A fence in that condition does not exclude kangaroos. It is very likely that the deferral contributed to the occurrence by leaving open the route through which the two kangaroos reached the runway strip. The aerodrome operator's own register had documented both the breach and animals passing through it since 2023.

The aerodrome operator has taken the following proactive safety action, which is in progress:

- It funded works register item WR-221 from its 2026 contingency and programmed replacement of the 140 m northern boundary fence section for 09/2026.
- It installed a temporary electric wire on 25/06/2026.

Evening service approved as a schedule change, route risk assessment not reopened

The operator introduced a scheduled evening rotation into Tarlton Springs as a schedule change and did not reopen the route risk assessment for the aerodrome, even though that assessment restricted operations there to daylight. It is virtually certain that this occurred, and likely that it contributed to the occurrence.

Route Risk Assessment RRA-041 issue 3 (11/2022) was the current assessment for Tarlton Springs. It carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy". CMR-2025-118, approved on 22/05/2025, introduced the first company operation into the aerodrome after last light, from 14/07/2025. The record assessed the change as a schedule change only, identified no new hazard, and recorded that no route risk assessment review was required. RRA-041 was not reopened and no later issue was produced.

This matters because the change placed company aircraft into an aerodrome after last light for the first time, against a restriction in the operator's own current documentation. The change process asked whether the change was new, not whether the existing assessment still held. As a result, the aerodrome's wildlife exposure and its absence from the Appendix 8A list of aerodromes requiring a runway inspection went unexamined before night operations began. A review might have addressed only lighting and fuel, the matters the restriction named. Because the approver could not be interviewed, the reasons for the assessment could not be established.

No safety action has been recorded in response to this safety issue.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A 8.14.2 required the pilot in command to request a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. Section 8.14.3 required nothing at other aerodromes. An aerodrome could be listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked a crew to report a sighting, and no other evidence could admit an aerodrome to the list. Tarlton Springs had never been among the nine listed aerodromes. The list was last reviewed on 03/03/2025, four months before the evening service began.

This was the operator's own defence against this kind of occurrence, and it was not armed at Tarlton Springs. The control itself was sound, but its trigger could be met only by the event it existed to prevent. The aerodrome's sighting log held 11 entries of kangaroos on the sealed surface. None of them could place the aerodrome on the list, because they were not strikes and were not reported to the company.

On 26/06/2026 the Head of Flying Operations amended Appendix 8A by temporary revision TR-2026-04 to list Tarlton Springs. The 8.14.2 inspection requirement therefore applies to every departure there after last light, pending a permanent change. This action is closed.

The investigation recommended on 02/07/2026 that the operator:

- replace the Appendix 8A listing criterion so that an aerodrome is listed on its wildlife exposure rather than on a reported strike
- add a route risk assessment trigger for any aerodrome first operated into after last light.

The present criterion can only be met by the occurrence the control exists to prevent, so no aerodrome can be listed before a strike has happened there. A response was due by 30/09/2026, and the recommendation is open.

Rejected take-off training covers aircraft system failures below V1 only

It was extremely likely that the operator's recurrent rejected take-off training prepared crews only for aircraft system events, and gave no guidance or practice for an obstruction on the runway at any speed. Recurrent training syllabus TRG-SYL-07 module 4 contained four scripted scenarios: an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt and an engine failure at V1 plus 5 kt with the take-off continued. No crew had been presented with an external obstruction scenario in the five training cycles on file, and 214 crew completed the module in the current cycle alone. Operations Manual Part B 4.6 was consistent with the syllabus. It required the take-off to be continued at or above V1 and said nothing about an obstruction on the runway.

The issue mattered because it applied across the fleet. Every crew flying for the operator would meet an obstruction on the runway without a briefed or practised frame for the decision. In this occurrence, the pilot in command made the decision to reject within about 2 seconds, and both flight crew stated independently that every stop or go scenario they had been given was an engine event below V1. It was likely that the training gap contributed to the reject. The pilot in command also stated that he knew the V1 rule and departed from it deliberately because he judged the alternative worse. On that reading, the gap explains the conditions under which the decision was made but does not excuse it, and it remains a finding about the fleet.

No safety action has been recorded in response to this safety issue.

Cabin floor proximity emergency lighting partially unserviceable

The cabin floor proximity emergency lighting on VH-SBQ was virtually certainly partially unserviceable. Neither the pre-flight test nor the 24-month functional test interval could reliably detect the defect.

The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed. The fault was traced to a failed common ballast. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but does not illuminate or monitor the floor segments. The crew or a line engineer therefore could not have detected the defect without a dedicated functional test. The last such test had been carried out 11 months before the occurrence, within the 24-month programme interval.

The defect did not contribute to the overrun or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and the disembarkation by the forward airstair took 4 minutes 8 seconds without incident. The defect very likely increased risk nonetheless. If the circumstances had included a fire indication, smoke or a night evacuation over the wing, two dark segments in the aft cabin could have been the difference between an orderly exit and a blocked one. The daily test cannot detect a fault in the floor segments, and the functional test interval is 24 months. As a result, the true state of these segments across the fleet is unknown, and the same ballast is fitted across the type.

Safety action

No safety action has been recorded in relation to this safety issue.

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